

Extra questions (Year 10+ 10A)

Exercise 3.1 Expressions and polynomials

1 Simplify if possible.

a $4a^2 \times 3a + 8a^3 - 9ba + 2ab$

c $4d \times 2edf - 9d^2ef + d^2fe$

e $5s \times 3t \times 2u + s^2 \times 6tu$

g $18u \div 3 \times 2u + 2u^2 \times 7u - 8u^3$

i $6xy + 4x \times 2y^2 - x \times y \times y$

k $8k - 2k^3 + 7k \times k^2 - 5k \times -k^2$

b $d^2 + 2d^3 + d \times 5d - 9d$

d $3pq \times 3pq - 5p^2q + 8qp$

f $2x \times y + 3y \times x + 4x \times y$

h $6g \times h^2 - 8g \times 2h \times 3h$

j $-6q \times 4r + -3r \times -7r - qr$

l $9vw^2 - 7vw + w^2v + 5v^2w$

2 Simplify each of the following.

a $8a^3gp^2 \times a^2g^3p^3$

b $8ckm \times 9ck^2m^3$

c $4b^3p^2t^2 \times b^2pt^3$

d $5b^3gq \times 2bg^3$

e $4cn^3x^3 \times cn^3x^2$

f $2c^3g^2 \times 5cg^3m$

g $8emv \times 7em^2v^3$

h $b^3q^2v^3 \times 4b^3q^2v$

i $4g^2h^3m \times ghm^2$

j $6c^2p^2t^3 \times 5cpt^3$

3 Simplify each of the following.

a $18ak^3q \div 27a^2k^2q$

b $21h^2mp^2 \div 18h^2p^3$

c $20a^2e^3g^2 \div 50a^2e^2g$

d $40g^3r^3t \div 24g^2r^2t^2$

e $5nt^2v^2 \div 6n^2tv^2$

f $30r^2v^3 \div 42rv^3y$

g $30a^2g^2t^3 \div 35a^3gt^3$

h $32ak^2n^3 \div 4a^2kn^2$

i $40aq^3r^2 \div 5aq^3r^2$

j $24b^2ek \div 9be^2k^3$

4 Find the value the nominated variable in each of the following formulas.

a u if $s = ut + \frac{1}{2}at^2$ and $s = 46.5$, $a = 5$ and $t = 3$

b R if $I = \frac{E - V}{R}$ and $I = 9$, $E = 18.7$ and $V = 5.2$

c g if $p = ghr + B$ and $p = 841$, $h = 35$, $r = 2.4$ and $B = 17.8$

d u if $v = \frac{w - u}{gr}$ and $v = 10$, $w = 251.8$, $g = 9.8$ and $r = 2.5$

e a if $d = 7\sqrt{\frac{h}{a}}$ and $d = 56$ and $h = 32$

f h if $B = \sqrt{2gh}$ and $B = 22$ and $g = 10$

g c if $k = m(12 - ct)$ and $k = 32$, $m = 8$ and $t = 2$

h b if $a^2 = b^2 + c^2$ and $a = 26$ and $c = 10$

i v if $s = \frac{(u + v)t}{2}$ and $s = 11.7$, $u = 1.4$ and $t = 2.6$

j t if $v = u + at$ and $u = 12$, $a = 4$ and $v = 20$.

5 Simplify each of the following.

a $24a^9n^6r^9 \div 4a^{12}n^5r^8$

b $1e^6k^8 \times 8e^{10}kp^5$

c $8c^8e^2p^{11} \times 9c^{10}e^8p^{11}$

d $90c^2e^{10}m^{10} \div 50c^{10}e^7$

e $45c^3r^7v \div 35c^2r^{11}v$

f $5k^{11}q^2r^{10} \times 3k^6qr^6$

g $5a^{10}r^{12}v^5 \times 2a^2r^9v^9$

h $3a^2g^{11} \times 4a^4g^3t^5$

i $4a^{11}b^{12} \times 5a^8b^{12}k$

j $15k^3p^3v^6 \div 9k^8p^{11}v^9$

6 Simplify each of the following.

a $-36ab^{-4}w^{-4} \div 9a^2b^5w^4$

b $1e^4k^2r^{-4} \times 3e^{-1}k^5r^3$

c $50b^{-1}n^{-1} \div 40bn^3v$

d $-8g^5h^{-4}p^4 \div 2g^{-2}h^5p$

e $40c^5r^3t^4 \div -10c^3r^5t^5$

f $-18a^3c^4m^2 \div 24a^{-2}c^4m^{-2}$

g $25a^4b^5k^5 \div 20a^2bk^3$

h $-9apw^4 \div -3a^4pw^{-4}$

i $15b^{-2}h^{-3}k \div -3b^4h^{-5}k^{-4}$

j $25a^{-4}e^{-3}m \div 10a^{-3}e^{-4}m^4$

7 Transpose the following formulas so that the letter shown in brackets is the subject.

a $fg = 4t - 7y$ (y)

b $\frac{r}{4} = \frac{d}{2p}$ (d)

c $k = \frac{4r + px}{3d}$ (r)

d $y = \frac{h - 5}{p - k}$ (h)

e $x = 5(g - mn)$ (g)

f $A = a(5 - 4bc)$ (b)

8 State the leading term and degree of any of the following that are polynomials.

a $C(y) = 2y^3 + 8y^2 + 4y + 2$

b $R(x) = x^2 - \sqrt{x} + 5$

c $Q(z) = 4z^4 - 9z^3 + 4z^2$

d $D(a) = a^4 - a^7 - 7a^3 - 2$

e $B(m) = 3m^2 + 8m^4 - 1$

f $T(y) = 9y - 8y^3 - 3$

g $H(v) = 9v^7 - 9v^6 + \frac{1}{v - 4}$

h $S(n) = 7n^5 - 9n^{-3} - 9n + 3$

i $R(e) = 8e^2 - e + 5$

j $Q(v) = 6v^3 - v - 3v^5$

9 Simplify each of the following.

a $-7a^2g^{-2}p^2w^{-3} \times 10a^{-3}g^{-1}p^{-2}w^2$

b $-12b^{-3}r^3ux^{-2} \times -11b^{-1}r^{-1}u^{-2}x^{-3}$

c $9b^2r^{-2}v^{-3}z^{-1} \times -13b^{-3}r^{-2}v^3z$

d $10ab^{-2}h^{-2}y^{-3} \times -13a^{-1}b^{-3}h^{-1}y^{-1}$

e $-3nt^3w^3y^2 \times -10n^{-1}t^2w^{-2}y$

f $-14e^3r^3v^{-2}z^3 \times 13e^{-2}r^2v^{-2}z$

g $30b^{-1}c^{-2}m^2z \div 27b^{-3}m^3z^{-1}$

h $13g^{-3}h^3mw^2 \times -12g^{-2}h$

i $40a^{-1}b^{-2}v^2 \div -72a^{-2}b^2gv^{-3}$

j $28ak^3wz^{-3} \div 2ak^{-1}wz^2$

k $-90b^{-2}k^3y^{-1}z \div -50b^{-3}k^{-1}y^{-2}z^{-1}$

l $84k^2q^{-2}tu^{-1} \div 35kq^{-2}tu^2$

m $-9b^3m^3t^{-3} \times 4bm^{-3}tx^2$

n $6a^{-1}h^3z^{-1} \div -26a^3h^{-2}r^{-3}z^{-3}$

o $-65a^{-2}b^3e^2k^3 \div -45a^{-1}b^2e^{-3}k^2$

p $-12a^{-3}e^3r^2 \times 7a^{-1}em^{-1}r^{-3}$

q $ce^2wx^{-1} \times -14c^{-1}e^{-1}w^{-2}x^{-1}$

r $-16bu^2v^{-1}w^2 \div -8b^{-2}u^{-1}v^2w^{-2}$

s $14h^2q^3u^{-1}x^{-1} \times 13hq^3u^{-2}x^{-2}$

t $-14am^{-1} \times 3a^{-3}m^{-1}n^3z^2$

u $-6a^{-1}g^{-2}h^3u^{-3} \times 7a^3g^3h^{-1}u^{-1}$

v $112e^{-2}g^{-3}h^2w^3 \div -24e^3g^3h^3w^{-1}$

w $7am^{-3}qt^{-2} \div 12a^{-2}m^2q^2t^{-3}$

x $-5a^{-2}p^{-1}q^{-1} \div -6a^3p^{-3}q^{-1}x^{-1}$

y $-10a^2b^{-3}m^{-1}u^{-3} \times -11a^{-2}b^{-3}m$

z $-12r^{-1}u^{-2}w^3x^{-3} \div -30r^{-3}u^2w^3x$

10 Simplify each of the following.

a $\frac{6cw - 4cw}{5w^2x^2 - 2w^2x^2} \times (4wx^2 + 3wx^2)$

b $\frac{3x + x}{2ey + 3ey} \times \frac{7ey - 5ey}{5ex - 2ex}$

c $\frac{9yn - 7yn}{3y^2 + 5y^2} \times \frac{9nvy + 13nvy}{10ny - 7ny}$

d $\frac{10mr - 8mr}{10k^2 + 2k^2} \times \frac{12hk + 8hk}{10m - 9m}$

e $\frac{3eh + 5he}{3ge + 2eg} \times \frac{6ge^2 - 5ge^2}{3eh^2 - 2eh^2}$

f $\frac{9b^2h - 4b^2h}{7mp - 5mp} \times \frac{7ph - 4ph}{3mhb + 4mhb}$

11 $P(x) = 4x^5 - 9x^2 - 8x - 4$, $Q(x) = 7x^3 - 9x + 4$ and $R(x) = 3x^4 - 4x^3 + x^2 + 7$. Find each of the following.

a $P(1)$

b $P(-2)$

c $P(3)$

d $P(-2)$

e $P(4)$

f $P(2)$

g $Q(-1)$

h $Q(-3)$

i $Q(2)$

j $Q(-2)$

k $Q(-5)$

l $Q(5)$

m $R(-2)$

n $R(-3)$

o $R(3)$

p $R(-1)$

q $R(4)$

r $R(-4)$